

Figure 13.1: (a) 3D view of chloroplast (b) Sectional view of chloroplast

A sac like membranous system called **thylakoid** or **lamellae** is present in stroma and they are arranged one above the other forming a stack of coin like structure called **granum** (plural grana). Each chloroplast contains 40 to 80 grana and each granum consists of 5 to 30 thylakoids.

Thylakoids found in granum are called grana lamellae and in stroma are called stroma lamellae. Thylakoid disc size is 0.25 to 0.8 micron in diameter. A thinner lamella called Fret membrane connects grana. Pigment system I is located on outer thylakoid membrane facing stroma and Pigment system II is located on inner membrane facing lumen of thylakoid. Grana lamellae have both PS I and PS II whereas stroma lamellae have only PS I. Chloroplast contains 30–35 Proteins, 20–30% phospholipids, 5–10% chlorophyll, 4–5% Carotenoids, 70S ribosomes, circular DNA and starch grains. Inner surface of lamellar membrane consists of small spherical structure called as **Quantasomes**. Presence of 70S ribosome and DNA gives them status of semi-autonomy and proves endosymbiotic hypothesis which says chloroplast evolved from bacteria. Thylakoid contains pigment systems which produces ATP and NADPH + H^+ using solar energy. Stroma contains enzyme which reduces carbon di oxide into carbohydrates. In Cyanobacteria thylakoid lies freely in cytoplasm without envelope (Figure 13.1).

13.2 Photosynthetic Pigments

A photosynthetic pigment is a pigment that is present in chloroplasts or photosynthetic bacteria which captures the light energy necessary for photosynthesis (Table 13.1).

Table 13.1: Types of Photosynthetic pigments

Chlorophyll	Carotenoids	Phycobilins
1. Chlorophyll 'a' ($C_{55}H_{72}O_5N_4Mg$) – Green plants and Cyanobacteria	1. Carotene ($C_{40}H_{56}$) – Lycopene (Red)	1. Phycocyanin – Cyanobacteria
2. Chlorophyll 'b' ($C_{55}H_{70}O_6N_4Mg$) – Green algae and all higher plants	2. Xanthophyll ($C_{40}H_{56}O_2$ Yellow colour –Violaxanthin, Fuco-xanthin (Brown Algae) and Lutein	2. Phycoerythrin – Red Algae
3. Chlorophyll 'c' ($C_{55}H_{32}O_5N_4Mg$) – Dinoflagellates, Diatoms and Brown Algae		
4. Chlorophyll 'd' – Red Algae		
5. Chlorophyll 'e' – Xathophycean Algae		
6. Bacteriochlorophyll 'a'		
7. Bacteriochlorophyll 'b'		
8. Chlorobium Chlorophyll 650		
9. Chlorobium Chlorophyll 666		

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